Olist E-Commerce

Data Analysis

Sales, Customer, and Delivery Performance

Department: Computer Science – BINUS University

Project Overview

This project focuses on analyzing the Olist Brazilian E-Commerce Public Dataset to evaluate business performance across sales, customer behavior, and delivery operations.

The objective is to transform raw e-commerce data into meaningful insights that can help identify business opportunities and support data-driven decision-making.

Dataset

Dataset	Description	Records
Orders	Order status & timestamps	99.441
Order Items	Products, prices & freight values	112.650
Order Payments	Payment methods & transaction values	103.886
Customers	Customer identifiers & locations	99.441
Products	Product categories & attributes	32.951

Observation Period : 2016 - 2018

Total Order Records : 99.441

Main Datasets Used : 5

Analysis Areas : 3

The dataset consists of multiple interconnected tables containing information about orders, customers, products, payments, sellers, and delivery details. These datasets provide a comprehensive view of the e-commerce transaction process, from purchase to delivery.

Business Questions

Defining the key business questions to guide the analysis and uncover opportunities for business improvement.

1. Sales Performance Analysis

- How does revenue change over time ?
- Which product categories generate the highest revenue ?
- What is the Average Order Value (AOV) ?

2. Customer Behavior Analysis

- How many unique customers have made purchases ?
- What percentage of customers make repeat purchases ?
- Which states have the highest number of customers ?

3. Delivery & Operations Analysis

- What percentage of orders are delivered on time versus late ?
- What is the average delivery duration ?
- How does delivery performance change over time ?

Tools

Python

Data preprocessing, cleaning, and exploratory data analysis (EDA).

PostgreSQL

Database management and business data querying.

Power BI

Interactive dashboard development and KPI calculations.

Workflow

Data Understanding

- Identified the number of rows and columns across each dataset.
- Checked for missing values, duplicate records, and data types.
- Explored relationships between orders, customers, products, and payments.

Data Cleaning

- Converted timestamp columns into appropriate datetime formats.
- Handled missing values in product categories.
- Identified inconsistent timestamps and potential data anomalies.
- Validated duplicate records and referential integrity.

Exploratory Data Analysis

- Analyzed revenue trends and order performance over time.
- Identified top-performing product categories.
- Examined customer purchasing behavior and repeat customer rates.
- Evaluated delivery duration and on-time delivery performance.

SQL Analysis

- Imported cleaned datasets into a PostgreSQL database.
- Created and managed relational database tables.
- Used SQL JOINs to combine data from multiple tables.
- Applied aggregate functions such as SUM, COUNT, and AVG.

Data Visualization

- Developed KPI cards to highlight key business metrics.
- Visualized monthly revenue trends and product category performance.
- Evaluated delivery and operational performance.
- Designed a three-page interactive dashboard to present key findings.

Visualization

Olist - Sales Performance Analysis

Tracking revenue trends, order performance, and product sales.

Total Orders

96K

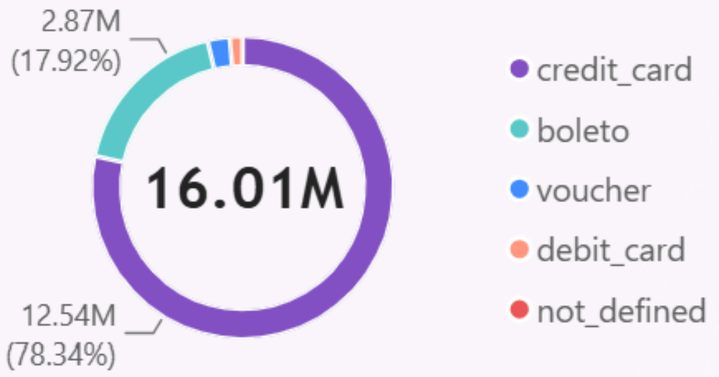
Total Revenue

13.22M

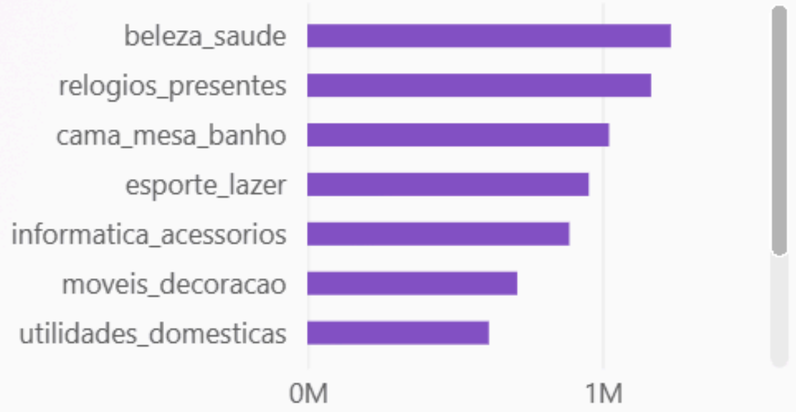
Average Order Value

137.04

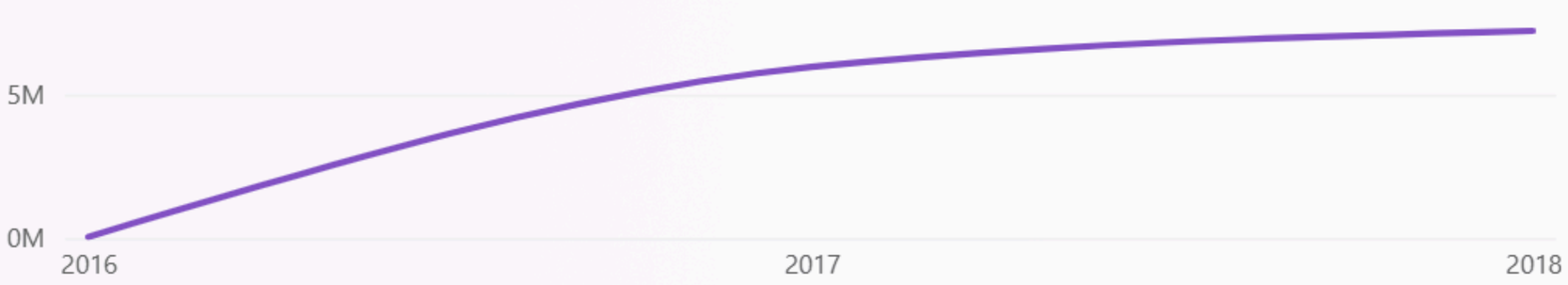
Sum of payment_value
by payment_type



Total Revenue
by product_category_name



Total Revenue
by Year



Insights

Product Category Performance

Beauty & Health (beleza_saude) generated the highest revenue at approximately R$1.23 million.

Watches & Gifts (relogios_presentes) ranked second with approximately R$1.17 million.

Bed, Bath & Table (cama_mesa_banho) ranked third with approximately R$1.02 million.

Revenue Performance

The analysis recorded approximately R$13.22 million in total revenue from 96,478 delivered orders, with an average order value of R$137.04.

Business Recommendations

Prioritize inventory planning for high-revenue product categories to maintain product availability.

Investigate seasonal purchasing patterns to better understand monthly revenue fluctuations.

Explore promotional strategies to improve the performance of lower-revenue categories.

Olist - Customer Behavior Analysis

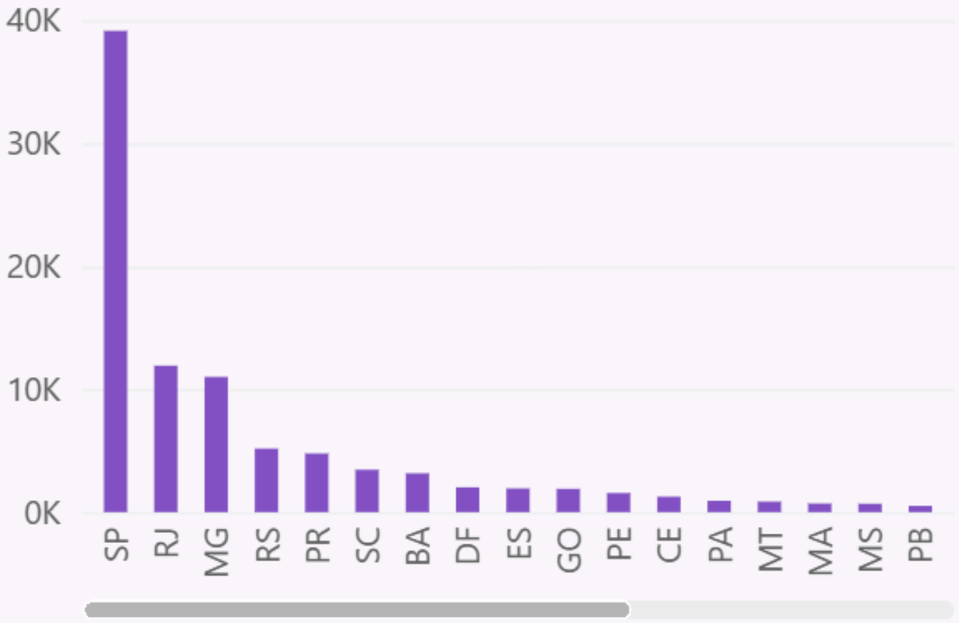
Analyzing customer behavior, purchasing patterns, and geographic distribution to understand the Olist customer base.

Unique Customer
93K

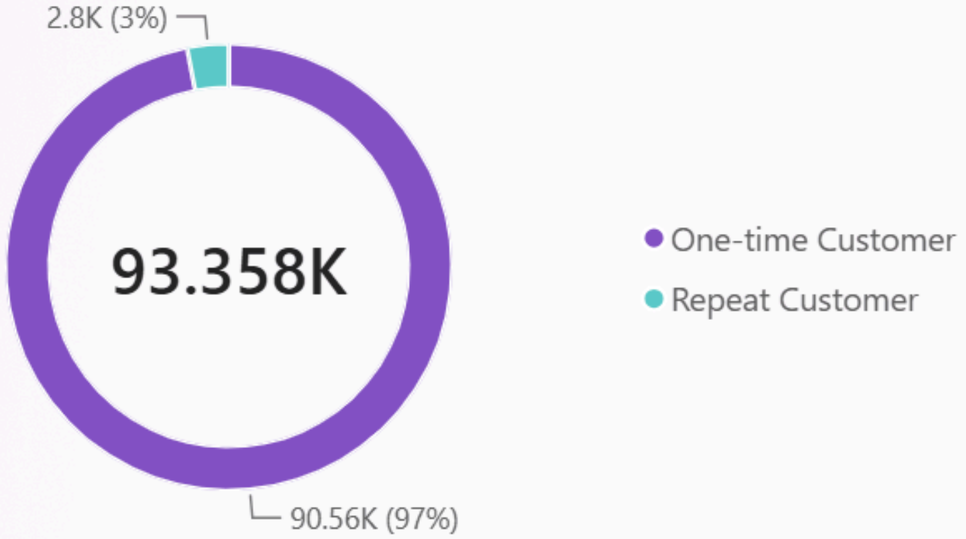
Repeat Customers
3K

Repeat Customers Rate
0.03

Unique Customers by State
by customer_state



Count of customer_unique_id
by Customer Type



Insights

Customer Purchasing Behavior

The analysis identified 93,358 unique customers with delivered orders.

90,557 customers (approximately 97%) made only one purchase.

Only 2,801 customers (approximately 3%) made repeat purchases.

Geographic Customer Distributions

Customer distribution across states provides insights into Olist's geographic market reach. States with higher customer concentrations may represent important markets for targeted marketing campaigns, customer engagement initiatives, and operational planning.

Business Recommendations

Introduce loyalty programs and incentives to encourage customers to make repeat purchases.

Develop personalized promotions based on customer purchasing behavior.

Focus marketing campaigns on states with strong customer demand while exploring opportunities in less-penetrated regions.

Delivery & Operations Analysis

Evaluating delivery performance, shipping times, and order fulfillment to identify operational inefficiencies.

Average Delivery Days

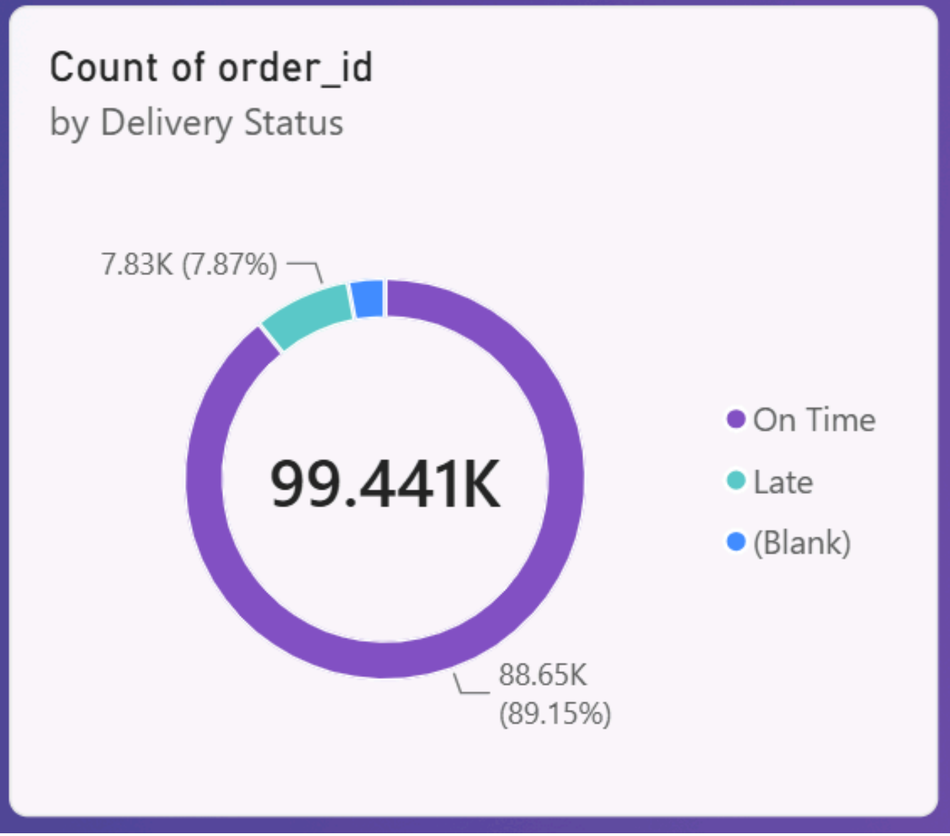
12.56

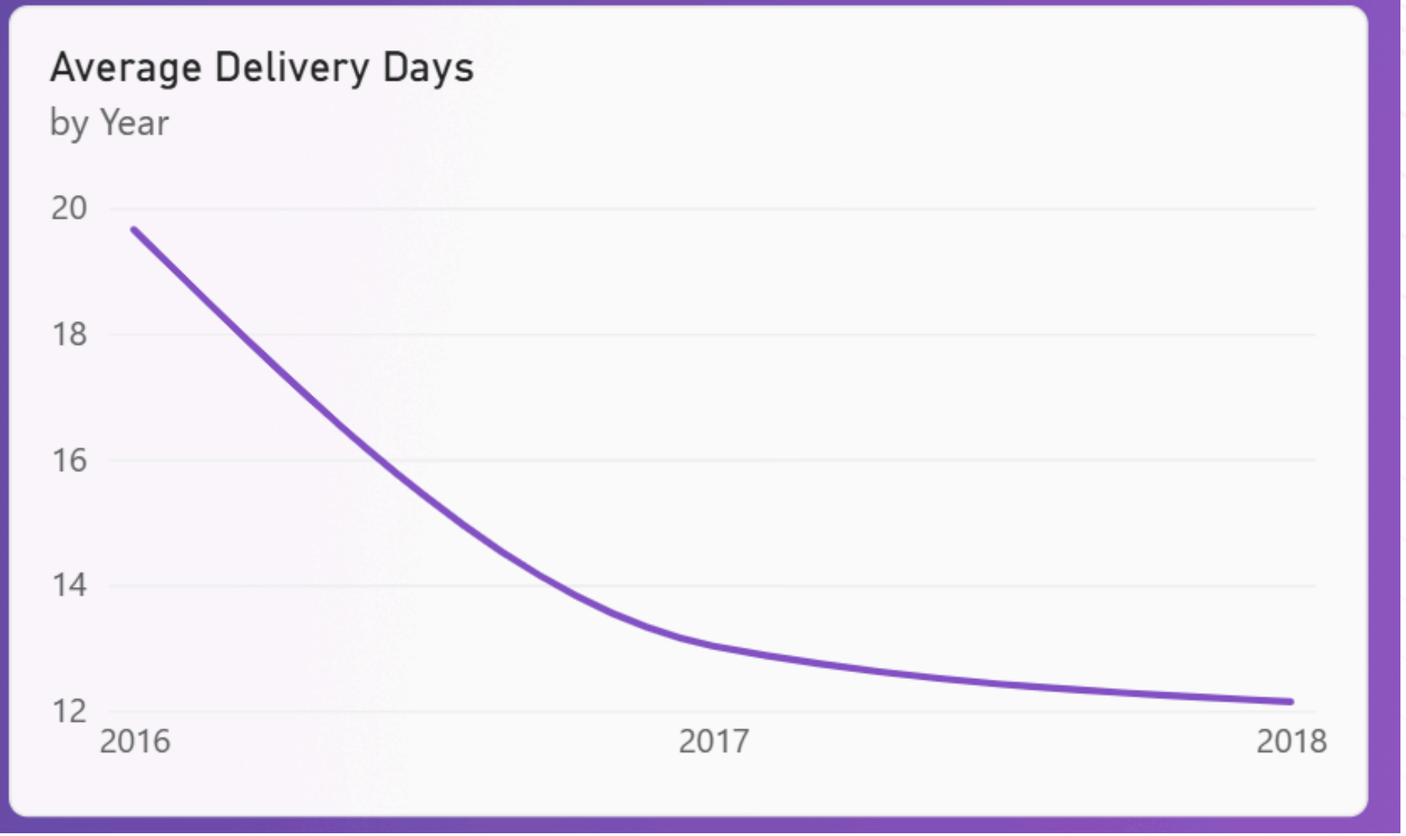
On-Time Delivery Rate

0.92

Late Delivery Rate

0.08





Insights

Delivery Purchase Analysis

91.89% of eligible orders were delivered on time.

8.11% of eligible orders experienced delivery delays and the average delivery duration was approximately 12.56 days.

Late deliveries experienced an average delay of approximately 9.55 days beyond the estimated delivery date.

Delivery Duration Trends

Monitoring monthly delivery duration can help identify periods of operational inefficiency.

Understanding these variations may support better logistics planning, resource allocation, and delivery performance management.

Business Recommendations

Improve logistics coordination and resource allocation during periods of high order volume.

Review estimated delivery dates to ensure they reflect realistic shipping conditions.

Track on-time delivery rates and average delivery duration to measure operational improvements.

Conclusion

Overall Findings

Strong Revenue Contribution from Key Product Categories

Olist generated approximately R$13.22 million in revenue from 96,478 delivered orders. Beauty & Health, Watches & Gifts, and Bed, Bath & Table were the top revenue-generating categories, highlighting their contribution to overall sales performance.

Low Repeat Purchase Rate

Only 3.00% of unique customers made repeat purchases within the observed dataset, while approximately 97% made only one purchase. This indicates an opportunity to strengthen customer retention and encourage repeat transactions.

Generally Reliable Delivery Performance

Olist achieved an on-time delivery rate of 91.89%, with an average delivery duration of 12.56 days. However, 8.11% of eligible orders were delivered late, indicating room for improvement in logistics and delivery management.

Business Recommendations

Optimize High-Performing Product Categories

Prioritize inventory availability and targeted marketing for top-performing categories while exploring strategies to improve the performance of lower-revenue categories.

Strengthen Customer Retention

Introduce loyalty programs, personalized promotions, and post-purchase engagement initiatives to encourage customers to make repeat purchases.

Improve Delivery Monitoring

Monitor delayed orders, evaluate logistics performance, and review estimated delivery dates to improve delivery reliability and the customer experience.

Monitor Business Performance Regularly

Use sales, customer, and delivery KPIs to track business performance, identify emerging trends, and support data-driven decisions.

Thank You

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https://github.com/discaaa/Olist_Analysis